

Stationary Maxwell-Jüttner Distribution functions: Detailed derivation.

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ABSTRACT

We present a detailed derivation of some calculations related to the manifestly covariant Maxwell-Jüttner distribution function (hereafter MJ DF) of a jet as seen in the laboratory frame where the jet is propagating with (dimensionless) speed β_J . We also consider the case of the modified MJ DF (MMJ) of a jet in thermal equilibrium with an electron synchrotron radiation field. Finally, we compute some moments of these DF which we use to derive physical properties of the jet-cloud interactions.

Keywords: Numerical techniques, Asymptotic expansions

1 ASYMPTOTIC EXPANSION OF MOMENTS OF MAXWELL-JÜTTNER.

Our starting point is the manifestly covariant Maxwell-Jüttner (hereafter MJ) distribution function (DF) introduced by Chacon-Acosta et al. (Chacón-Acosta et al., 2010, eq. (33)):

$$f(p) d^3\mathbf{p} = A \exp(-\Theta^\nu P_\nu) d^3P \quad (1)$$

where: $\Theta^\nu = (c/k_B T) V^\nu$ and the dimensionless four-velocity velocity is defined as: $V_J^\nu \equiv (\gamma_J, \gamma_J \beta_J)$, $\gamma_J = (1 - |\beta_J|^2)^{-1/2}$ being the Lorentz factor corresponding to the jet's 3D spatial dimensionless velocity $\beta_J = \mathbf{V}_J/c$. We introduce a dimensionless 4-momentum $p^\mu \equiv P^\mu/mc$, so that: $\Theta^\nu P_\nu = \lambda \gamma_J \gamma (1 - \beta_J \cdot \beta)$, $\lambda = mc^2/k_B T$.

We now consider a reference frame with one axis parallel to the direction of the jet. Assuming cylindrical coordinates in momentum space, the reduced momentum is expressed as: $p^i \equiv P^i/(mc) = (p_\perp \cos \phi, p_\perp \sin \phi, p_\parallel)$, and one easily verifies that the volume element can be expressed as: $d^3p \rightarrow p_\perp dp_\perp dp_\parallel d\phi \rightarrow \gamma d\gamma dp_\parallel d\phi$, where we have used the relation: $\gamma^2 = 1 + p_\parallel^2 + p_\perp^2$, obtained starting from the 4-momentum conservation relation: $P^\mu P_\mu = (m^2 c^4) \gamma^2 - P_\perp^2 - P_\parallel^2 = m^2 c^4$.

We now adopt coordinates $(\gamma, p_\parallel, \phi)$ instead of $(p_\perp, p_\parallel, \phi)$, and thus the volume element in the new phase space becomes: $d^3P \rightarrow (mc)^2 \gamma d\gamma dp_\parallel d\phi$. With these new coordinates we then obtain: $\Theta^\nu P_\nu = \lambda \gamma_J \gamma (1 - \beta_J \cdot \beta) = \lambda \gamma_J \gamma - \lambda \beta_J \gamma_J \gamma p_\parallel$, because: $\gamma \beta_\parallel = P_\parallel/m_e c = p_\parallel$. Thus, with these coordinates the manifestly covariant Maxwell-Jüttner distribution function in a reference frame where the jet is seen to be moving at speed $\mathbf{V}_J$ is given by:

$$f_{MJ}(\gamma, p_\parallel, \phi) d^3\mathbf{p} = A \exp(-\Theta^\nu P_\nu) d^3P = A \exp(-\lambda \gamma_J \gamma + \lambda \beta_J \gamma_J \gamma p_\parallel) \cdot \gamma d\gamma dp_\parallel d\phi \quad (2)$$

Here we have assumed f_{MJ} to be axisymmetric ($\partial/\partial\phi = 0$) and introduced the bulk dimensionless velocity $\beta_J = V_J/c$, the Lorentz factor $\gamma_J = (1 - \beta_J^2)^{-1/2}$ and the only parameter entering the MJ DF: $\lambda = 1/\tau = (k_B T/m_e c^2) \simeq T/(5.93 \times 10^9 K)$, where T is the equilibrium temperature in the fluid's rest frame. This DF is manifestly anisotropic.

The normalization factor A in these new momentum coordinates can easily be computed starting from the expression given by Chacón-Acosta et al. (2010, eq. (33)): $A = A_{CA2010} \cdot (mc)^2 \cdot 2\pi$, thus obtaining:

$$A = \frac{N}{2(m_e c^2) \tau K_2(1/\tau)} \quad (3)$$

where N is the spatial density in the jet's rest frame.

1.0.1 Modified stationary Maxwell-Jüttner DF with synchrotron radiation

Ochs et al. (2024) have recently considered the evolution equation of the electron distribution function in presence of a synchrotron radiation field.

We start from the transport equation given by Ochs et al, eq. (2):

$$\frac{\partial f}{\partial t} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial \mathbf{u}} \cdot \left[\sqrt{g} \left(-\mathbf{\Gamma} f + \mathbf{D} \cdot \frac{\partial f}{\partial \mathbf{u}} \right) \right] \quad (4)$$

where $\mathbf{\Gamma}$ and $\mathbf{D}$ are the drag vector and the diffusion diadic tensor, respectively. Ochs et al. (2024) show that, for the case of a black-body radiation spectrum in the Rayleigh-Jeans limit one finds (Ochs et al., 2024, eq. 17):

$$\mathbf{\Gamma} = -v_{R0} \frac{\gamma_{\perp}^2}{\gamma} \mathbf{g}; \quad \mathbf{g} \equiv \begin{pmatrix} p_{\perp} \\ \frac{p_{\perp}^2}{\gamma_{\perp}^2} p_{\parallel} \end{pmatrix} \quad (5)$$

where we have defined:

$$v_{R0} = \frac{2}{3} \frac{e^2 \Omega_0^2}{mc^3}; \quad \gamma_{\perp}^2 = 1 + p_{\perp}^2. \quad (6)$$

and $\Omega_0 = eB/m_e c$ is the (nonrelativistic) cyclotron frequency. We can then usefully simplify the expression for the drag term above as: $\mathbf{\Gamma} = -v_{\mathbf{g}}$. In the following we will always consider the effective, momentum dependent synchrotron time scale defined by Ochs et al, (their eq.18):

$$v_R \equiv v_{R0} \frac{\gamma_{\perp}^2}{\gamma} \quad (7)$$

For the particular blackbody spectrum assumed here Ochs et al. find the following expression for the diadic diffusion tensor $\mathbf{D}$:

$$\mathbf{D}_{bb} = v_R \frac{T_{bb}}{mc^2} \frac{\gamma_{\perp}^2}{\gamma u_{\perp}^2} (\mathbf{G} + \Delta \mathbf{H}), \quad (8)$$

where:

$$\mathbf{G} = \begin{pmatrix} u_{\perp}^2 & \frac{u_{\perp}^3 u_{\parallel}}{\gamma_{\perp}^2} \\ \frac{u_{\perp}^3 u_{\parallel}}{\gamma_{\perp}^2} & \frac{u_{\perp}^4 u_{\parallel}^2}{\gamma_{\perp}^4} \end{pmatrix} = \mathbf{g} \mathbf{g} \quad (9)$$

$$\mathbf{H} = \begin{pmatrix} u_{\parallel}^2 & -u_{\perp} u_{\parallel} \\ -u_{\perp} u_{\parallel} & u_{\perp}^2 \end{pmatrix} = \mathbf{h} \mathbf{h}, \quad \mathbf{h} \equiv \begin{pmatrix} u_{\parallel} \\ -u_{\perp} \end{pmatrix} \quad (10)$$

We are now ready to search for solutions of the evolution equation, eq. 4. We will assume a *modified Maxwell-Jüttner* DF of the form:

$$f = f_{MJ} \cdot \lambda(p_{\perp}, p_{\parallel}) \quad (11)$$

where f_{MJ} is the manifestly covariant DF for a constant velocity jet of speed $\mathbf{V}_J$ (eq. 2) and we will look for stationary solutions ($\partial f / \partial t = 0$). First of all, from Ochs et al. (2024, eq. 7) we have: $\sqrt{g} = 2\pi p_{\perp}$, and:

$$\mathbf{D}_{bb} \cdot \frac{\partial f}{\partial \mathbf{p}} = \frac{v_R}{\tau_{BB}} \cdot \frac{\gamma_{\perp}^2}{\gamma u_{\perp}^2} \mathbf{g} \left(\mathbf{g} \cdot \frac{\partial f}{\partial \mathbf{p}} \right) \quad (12)$$

where we have defined: $\tau_{BB} = k_B T_{BB} / m_e c^2$.

We will now derive the partial derivatives of the DF f .

$$\frac{\partial f}{\partial p_{\perp}} = -\tau_e \gamma_J \frac{p_{\perp} \gamma}{f_{MJ}} \lambda + f_{MJ} \frac{\partial \lambda}{\partial p_{\perp}} \quad (13)$$

$$\frac{\partial f}{\partial p_{\parallel}} = \tau_e \beta_J \gamma_J f_{MJ} \lambda - \tau_e \gamma_J \frac{p_{\parallel}}{\gamma} f_{MJ} \lambda + f_{MJ} \frac{\partial \lambda}{\partial p_{\parallel}} \quad (14)$$

We now compute the term: $\mathbf{G} \cdot \partial f / \partial \mathbf{p}$:

$$\mathbf{G} \cdot \frac{\partial f}{\partial \mathbf{p}} = \left(-\tau_e \gamma_J \frac{p_\perp^2}{\gamma} f_{MJ} \lambda + p_\perp f_{MJ} \frac{\partial \lambda}{\partial p_\perp} \right) + \frac{p_\perp^2 p_\parallel}{\gamma_\perp^2} \left(-\tau_e \gamma_J \frac{p_\parallel}{\gamma} f_{MJ} \lambda + \tau_e \beta_J \gamma_J f_{MJ} \lambda + f_{MJ} \frac{\partial \lambda}{\partial p_\parallel} \right) \mathbf{g} \quad (15)$$

After rearranging some terms we obtain:

$$f_{MJ} \left(-\frac{\tau_e \gamma_J}{\gamma} \lambda p_\perp^2 \left[1 + \frac{p_\parallel^2}{\gamma_\perp^2} \right] + p_\perp \frac{\partial \lambda}{\partial p_\perp} + \frac{p_\perp^2 p_\parallel}{\gamma_\perp^2} \left[\tau_e \beta_J \gamma_J \lambda + \frac{\partial \lambda}{\partial p_\parallel} \right] \right) \mathbf{g} \quad (16)$$

Now, we have:

$$1 + \frac{p_\parallel^2}{\gamma_\perp^2} = \frac{1 + p_\perp^2 + p_\parallel^2}{\gamma_\perp^2} = \frac{\gamma^2}{\gamma_\perp^2}$$

and thus the first term in the round parentheses in eq. 16 becomes:

$$-\frac{\tau_e \gamma_J}{\gamma} \lambda p_\perp^2 \left[1 + \frac{p_\parallel^2}{\gamma_\perp^2} \right] = -\tau_e \gamma_J \lambda \cdot \gamma \frac{p_\perp^2}{\gamma_\perp^2}$$

After substituting in eq. 16 and having multiplied by the factor $v_R / \tau_{BB} \cdot (\gamma_\perp^2 / \gamma p_\perp^2)$ from eq. 8 we eventually obtain:

$$\mathbf{D}_{BB} \cdot \frac{\partial f}{\partial \mathbf{p}} = \frac{v_R}{\tau_{BB}} f_{MJ} \left(-\tau_e \gamma_J \lambda + \frac{\gamma_\perp^2}{\gamma p_\perp^2} p_\perp \frac{\partial \lambda}{\partial p_\perp} + \frac{p_\parallel}{\gamma} \left[\tau_e \beta_J \gamma_J \lambda + \frac{\partial \lambda}{\partial p_\parallel} \right] \right) \mathbf{g} \quad (17)$$

We now evaluate the r.h.s. of eq. 4:

$$-\mathbf{\Gamma} f + \mathbf{D}_{BB} \cdot \frac{\partial f}{\partial \mathbf{u}} = v_R f_{MJ} \left[\lambda - \frac{\tau_e \gamma_J}{\tau_{BB}} + \frac{1}{\gamma \tau_{BB}} \left\{ \frac{\gamma_\perp^2}{p_\perp} \frac{\partial \lambda}{\partial p_\perp} + \frac{p_\parallel}{\gamma} \left(\tau_e \gamma_J \beta_J \lambda + \frac{\partial \lambda}{\partial p_\parallel} \right) \right\} \right] \mathbf{g} \quad (18)$$

1.1 Stationary solutions

We now assume that the Rayleigh-Jeans synchrotron field is in equilibrium with the jet's electron component: $\tau_{BB} = \gamma_J \tau_e$, thus we see that eq. 18 reduces to:

$$\frac{\gamma_\perp^2}{p_\perp} \frac{\partial \lambda}{\partial p_\perp} + \frac{p_\parallel}{\gamma} \left(\tau_e \gamma_J \beta_J \lambda + \frac{\partial \lambda}{\partial p_\parallel} \right) = 0 \quad (19)$$

We will look for *separable* solutions of this equation, of the form: $\lambda(p_\perp, p_\parallel) = P(p_\parallel) Q(p_\perp)$, and by substitution we obtain:

$$\frac{\gamma_\perp^2}{p_\perp} P Q' + p_\parallel Q P' + (\tau_e \beta_J \gamma_J) p_\parallel P Q = 0$$

and after dividing by $\lambda = PQ$ and using $\gamma_{p_\perp}^2 = 1 + p_\perp^2$ we arrive at:

$$\frac{1 + p_\perp^2}{p_\perp} \frac{Q'}{Q} + p_\parallel \left(\frac{P'}{P} + \tau_e \beta_J \gamma_J \right) = 0$$

The two terms in this equation are now independent. the first only depends on $p_\perp$, while the second on $p_\parallel$: we can then put them separately equal to an arbitrary constant c and look for solutions of two separate equations:

$$\frac{1 + p_\perp^2}{p_\perp} \frac{Q'}{Q} = c, p_\parallel \left(\frac{P'}{P} + \tau_e \beta_J \gamma_J \right) = -c \quad (20)$$

By choosing $c = 1$ we easily verify that we arrive at the following solutions:

$$Q(p_{\perp}) = (1 + p_{\perp}^2)^{1/2} + Q(0), P(p_{\parallel}) = \frac{1}{p_{\parallel}} \exp[-(\tau_e \beta_J \gamma_J) p_{\parallel}] \quad (21)$$

Thus the multiplicative component of the modified MJ DF becomes:

$$\lambda(p_{\perp}, p_{\parallel}) = \frac{(1 + p_{\perp}^2)^{1/2}}{p_{\parallel}} \exp[-(\tau_e \beta_J \gamma_J) p_{\parallel}] \quad (22)$$

Eventually, the modified MJ DF thus becomes:

$$f = f_{MJ} \lambda = \frac{(1 + p_{\perp}^2)^{1/2}}{p_{\parallel}} \exp[-(\tau_e \gamma_J) \gamma] \quad (23)$$

because the exponential component of λ exactly cancels the anisotropic component of f_{MJ} .

Note the suppression of the high $p_{\parallel}$ tail of the modified DF w.r.t. the isotropic one. We thus see that when a synchrotron radiation field is present the DF is significantly modified, if the synchrotron field is in equilibrium with the electron DF. . . .

REFERENCES

- Chacón-Acosta, G., Dagdug, L., and Morales-Técotl, H. A. (2010). Manifestly covariant Jüttner distribution and equipartition theorem. *Phys. Rev. E*, 81(2):021126.
- Ochs, I. E., Mlodik, M. E., and Fisch, N. J. (2024). Electron tail suppression and effective collisionality due to synchrotron emission and absorption in mildly relativistic plasmas. *Physics of Plasmas*, 31(8):083303.